積分表

附录: 积分表

(一) 含有 $ax+b$ 的积分

1. $\displaystyle\int \frac{\mathrm{d}x}{ax+b} = \frac{1}{a}\ln|ax+b| + C.$
2. $\displaystyle\int (ax+b)^{\mu}\mathrm{d}x = \frac{1}{a(\mu+1)}(ax+b)^{\mu+1} + C(\mu \neq -1).$
3. $\displaystyle\int \frac{x}{ax+b}\,\mathrm{d}x = \frac{1}{a^2}(ax+b-b\ln|ax+b|) + C.$
4. $\displaystyle\int \frac{x^2}{ax+b}\,\mathrm{d}x = \frac{1}{a^3}\left[\frac{1}{2}(ax+b)^2 - 2b(ax+b) + b^2\ln|ax+b|\right] + C.$
5. $\displaystyle\int \frac{\mathrm{d}x}{x(ax+b)} = -\frac{1}{b}\ln\left|\frac{ax+b}{x}\right| + C.$
6. $\displaystyle\int \frac{\mathrm{d}x}{x^2(ax+b)} = -\frac{1}{bx} + \frac{a}{b^2}\ln\left|\frac{ax+b}{x}\right| + C.$
7. $\displaystyle\int \frac{x}{(ax+b)^2}\,\mathrm{d}x = \frac{1}{a^2}\left(\ln|ax+b| + \frac{b}{ax+b}\right) + C.$
8. $\displaystyle\int \frac{x^2}{(ax+b)^2}\,\mathrm{d}x = \frac{1}{a^3}\left(ax+b-2b\ln|ax+b| - \frac{b^2}{ax+b}\right) + C.$
9. $\displaystyle\int \frac{\mathrm{d}x}{x(ax+b)^2} = \frac{1}{b(ax+b)} - \frac{1}{b^2}\ln\left|\frac{ax+b}{x}\right| + C.$

(二) 含有 $\sqrt{ax+b}$ 的积分

10. $\displaystyle\int \sqrt{ax+b}\,\mathrm{d}x = \frac{2}{3a}\sqrt{(ax+b)^3} + C.$
11. $\displaystyle\int x\sqrt{ax+b}\,\mathrm{d}x = \frac{2}{15a^2}(3ax-2b)\sqrt{(ax+b)^3} + C.$
12. $\displaystyle\int x^2\sqrt{ax+b}\,\mathrm{d}x = \frac{2}{105a^3}\left(15a^2x^2 - 12abx + 8b^2\right)\sqrt{(ax+b)^3} + C.$
13. $\displaystyle\int \frac{x}{\sqrt{ax+b}}\,\mathrm{d}x = \frac{2}{3a^2}(ax-2b)\sqrt{ax+b} + C.$
14. $\displaystyle\int \frac{x^2}{\sqrt{ax+b}}\,\mathrm{d}x = \frac{2}{15a^3}\left(3a^2x^2 - 4abx + 8b^2\right)\sqrt{ax+b} + C.$
15. $\displaystyle\int \frac{\mathrm{d}x}{x\sqrt{ax+b}} = \begin{cases} \frac{1}{\sqrt{b}}\ln\left|\frac{\sqrt{ax+b}-\sqrt{b}}{\sqrt{ax+b}+\sqrt{b}}\right| + C & (b>0), \\ \frac{2}{\sqrt{-b}}\arctan\sqrt{\frac{ax+b}{-b}} + C & (b<0). \end{cases}$
16. $\displaystyle\int \frac{\mathrm{d}x}{x^2\sqrt{ax+b}} = -\frac{\sqrt{ax+b}}{bx} - \frac{a}{2b}\int \frac{\mathrm{d}x}{x\sqrt{ax+b}}.$
17. $\displaystyle\int \frac{\sqrt{ax+b}}{x}\,\mathrm{d}x = 2\sqrt{ax+b} + b\int \frac{\mathrm{d}x}{x\sqrt{ax+b}}.$
18. $\displaystyle\int \frac{\sqrt{ax+b}}{x^2}\,\mathrm{d}x = -\frac{\sqrt{ax+b}}{x} + \frac{a}{2}\int \frac{\mathrm{d}x}{x\sqrt{ax+b}}.$

(三) 含有 $x^2 \pm a^2$ 的积分

19. $\int \frac{\mathrm{d}x}{x^2+a^2} = \frac{1}{a}\arctan\frac{x}{a} + C.$

20. $\int \frac{\mathrm{d}x}{(x^2+a^2)^n} = \frac{x}{2(n-1)a^2\,(x^2+a^2)^{n-1}} + \frac{2n-3}{2(n-1)a^2}\int \frac{\mathrm{d}x}{(x^2+a^2)^{n-1}}.$

21. $\int \frac{\mathrm{d}x}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C.$

(四) 含有 $ax^2+b(a>0)$ 的积分

22. $\int \frac{\mathrm{d}x}{ax^2+b} = \begin{cases} \frac{1}{\sqrt{ab}}\arctan\sqrt{\frac{a}{b}}x + C & (b>0), \\ \frac{1}{2\sqrt{-ab}}\ln\left|\frac{\sqrt{a}x-\sqrt{-b}}{\sqrt{a}x+\sqrt{-b}}\right| + C & (b<0). \end{cases}$

23. $\int \frac{x}{ax^2+b}\,\mathrm{d}x = \frac{1}{2a}\ln\left|ax^2+b\right| + C.$

24. $\int \frac{x^2}{ax^2+b}\,\mathrm{d}x = \frac{x}{a} - \frac{b}{a}\int \frac{\mathrm{d}x}{ax^2+b}.$

25. $\int \frac{\mathrm{d}x}{x\,(ax^2+b)} = \frac{1}{2b}\ln\frac{x^2}{|ax^2+b|} + C.$

26. $\int \frac{\mathrm{d}x}{x^2\,(ax^2+b)} = -\frac{1}{bx} - \frac{a}{b}\int \frac{\mathrm{d}x}{ax^2+b}.$

27. $\int \frac{\mathrm{d}x}{x^3\,(ax^2+b)} = \frac{a}{2b^2}\ln\frac{\left|ax^2+b\right|}{x^2} - \frac{1}{2bx^2} + C.$

28. $\int \frac{\mathrm{d}x}{(ax^2+b)^2} = \frac{x}{2b\,(ax^2+b)} + \frac{1}{2b}\int \frac{\mathrm{d}x}{ax^2+b}.$

(五) 含有 $ax^2+bx+c(a>0)$ 的积分

29. $\int \frac{\mathrm{d}x}{ax^2+bx+c} = \begin{cases} \frac{2}{\sqrt{4ac-b^2}}\arctan\frac{2ax+b}{\sqrt{4ac-b^2}} + C & \left(b^2<4ac\right), \\ \frac{1}{\sqrt{b^2-4ac}}\ln\left|\frac{2ax+b-\sqrt{b^2-4ac}}{2ax+b+\sqrt{b^2-4ac}}\right| + C & \left(b^2>4ac\right). \end{cases}$

30. $\int \frac{x}{ax^2+bx+c}\,\mathrm{d}x = \frac{1}{2a}\ln\left|ax^2+bx+c\right| - \frac{b}{2a}\int \frac{\mathrm{d}x}{ax^2+bx+c}.$

(六) 含有 $\sqrt{x^2+a^2}(a>0)$ 的积分

31. $\int \frac{\mathrm{d}x}{\sqrt{x^2+a^2}} = \operatorname{arsh}\frac{x}{a} + C_1 = \ln\left(x+\sqrt{x^2+a^2}\right) + C.$

32. $\int \frac{\mathrm{d}x}{\sqrt{(x^2+a^2)^3}} = \frac{x}{a^2\sqrt{x^2+a^2}} + C.$

33. $\int \frac{x}{\sqrt{x^2+a^2}}\,\mathrm{d}x = \sqrt{x^2+a^2} + C.$

34. $\int \frac{x}{\sqrt{(x^2+a^2)^3}}\,\mathrm{d}x = -\frac{1}{\sqrt{x^2+a^2}} + C.$

35. $\int \frac{x^2}{\sqrt{x^2+a^2}}\,\mathrm{d}x = \frac{x}{2}\sqrt{x^2+a^2} - \frac{a^2}{2}\ln\left(x+\sqrt{x^2+a^2}\right) + C.$

36. $\int \frac{x^2}{\sqrt{(x^2+a^2)^3}}\,\mathrm{d}x = -\frac{x}{\sqrt{x^2+a^2}} + \ln\left(x+\sqrt{x^2+a^2}\right) + C.$

37. $\int \frac{\mathrm{d}x}{x\sqrt{x^2+a^2}} = \frac{1}{a}\ln\frac{\sqrt{x^2+a^2}-a}{|x|} + C.$

38. $\int \frac{\mathrm{d}x}{x^2\sqrt{x^2+a^2}} = -\frac{\sqrt{x^2+a^2}}{a^2x} + C.$

39. $\int \sqrt{x^2+a^2}\,\mathrm{d}x = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\ln\left(x+\sqrt{x^2+a^2}\right) + C.$

40. $\int \sqrt{(x^2+a^2)^3}\,\mathrm{d}x = \frac{x}{8}\left(2x^2+5a^2\right)\sqrt{x^2+a^2} + \frac{3}{8}a^4\ln\left(x+\sqrt{x^2+a^2}\right) + C.$

41. $\int x\sqrt{x^2+a^2}\,\mathrm{d}x = \frac{1}{3}\sqrt{(x^2+a^2)^3} + C.$

42. $\int x^2\sqrt{x^2+a^2}\,\mathrm{d}x = \frac{x}{8}\left(2x^2+a^2\right)\sqrt{x^2+a^2} - \frac{a^4}{8}\ln\left(x+\sqrt{x^2+a^2}\right) + C.$

43. $\int \frac{\sqrt{x^2+a^2}}{x}\,\mathrm{d}x = \sqrt{x^2+a^2} + a\ln\frac{\sqrt{x^2+a^2}-a}{|x|} + C.$

44. $\int \frac{\sqrt{x^2+a^2}}{x^2}\,\mathrm{d}x = -\frac{\sqrt{x^2+a^2}}{x} + \ln\left(x+\sqrt{x^2+a^2}\right) + C.$

(七) 含有 $\sqrt{x^2-a^2}(a>0)$ 的积分

45. $\int \frac{\mathrm{d}x}{\sqrt{x^2-a^2}} = \frac{x}{|x|}\operatorname{arch}\frac{|x|}{a} + C_1 = \ln\left|x+\sqrt{x^2-a^2}\right| + C.$

46. $\int \frac{\mathrm{d}x}{\sqrt{(x^2-a^2)^x}} = -\frac{x}{a^2\sqrt{x^2-a^2}} + C.$

47. $\int \frac{x}{\sqrt{x^2-a^2}}\,\mathrm{d}x = \sqrt{x^2-a^2} + C.$

48. $\int \frac{x}{\sqrt{(x^2-a^2)^3}}\,\mathrm{d}x = -\frac{1}{\sqrt{x^2-a^2}} + C.$

49. $\int \frac{x^2}{\sqrt{x^2-a^2}}\,\mathrm{d}x = \frac{x}{2}\sqrt{x^2-a^2} + \frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right| + C.$

50. $\int \frac{x^2}{\sqrt{(x^2-a^2)^3}}\,\mathrm{d}x = -\frac{x}{\sqrt{x^2-a^2}} + \ln\left|x+\sqrt{x^2-a^2}\right| + C.$

51. $\int \frac{\mathrm{d}x}{x\sqrt{x^2-a^2}} = \frac{1}{a}\arccos\frac{a}{|x|} + C.$

52. $\int \frac{\mathrm{d}x}{x^2\sqrt{x^2-a^2}} = \frac{\sqrt{x^2-a^2}}{a^2x} + C.$

53. $\int \sqrt{x^2-a^2}\,\mathrm{d}x = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right| + C.$

54. $\int \sqrt{(x^2-a^2)^3}\,\mathrm{d}x = \frac{x}{8}\left(2x^2-5a^2\right)\sqrt{x^2-a^2} + \frac{3}{8}a^4\ln\left|x+\sqrt{x^2-a^2}\right| + C.$

55. $\int x\sqrt{x^2-a^2}\,\mathrm{d}x=\frac{1}{3}\sqrt{(x^2-a^2)^3}+C.$

56. $\int x^2\sqrt{x^2-a^2}\,\mathrm{d}x=\frac{x}{8}\left(2x^2-a^2\right)\sqrt{x^2-a^2}-\frac{a^4}{8}\ln\left|x+\sqrt{x^2-a^2}\right|+C.$

57. $\int\frac{\sqrt{x^2-a^2}}{x}\,\mathrm{d}x=\sqrt{x^2-a^2}-a\arccos\frac{a}{|x|}+C.$

58. $\int\frac{\sqrt{x^2-a^2}}{x^2}\,\mathrm{d}x=-\frac{\sqrt{x^2-a^2}}{x}+\ln\left|x+\sqrt{x^2-a^2}\right|+C.$

(八) 含有 $\sqrt{a^2-x^2}(a>0)$ 的积分

59. $\int\frac{\mathrm{d}x}{\sqrt{a^2-x^2}}=\arcsin\frac{x}{a}+C.$

60. $\int\frac{\mathrm{d}x}{\sqrt{(a^2-x^2)^3}}=\frac{x}{a^2\sqrt{a^2-x^2}}+C.$

61. $\int\frac{x}{\sqrt{a^2-x^2}}\,\mathrm{d}x=-\sqrt{a^2-x^2}+C.$

62. $\int\frac{x}{\sqrt{(a^2-x^2)^3}}\,\mathrm{d}x=\frac{1}{\sqrt{a^2-x^2}}+C.$

63. $\int\frac{x^2}{\sqrt{a^2-x^2}}\,\mathrm{d}x=-\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\arcsin\frac{x}{a}+C.$

64. $\int\frac{x^2}{\sqrt{(a^2-x^2)^3}}\,\mathrm{d}x=\frac{x}{\sqrt{a^2-x^2}}-\arcsin\frac{x}{a}+C.$

65. $\int\frac{\mathrm{d}x}{x\sqrt{a^2-x^2}}=\frac{1}{a}\ln\frac{a-\sqrt{a^2-x^2}}{|x|}+C.$

66. $\int\frac{\mathrm{d}x}{x^2\sqrt{a^2-x^2}}=-\frac{\sqrt{a^2-x^2}}{a^2x}+C.$

67. $\int\sqrt{a^2-x^2}\,\mathrm{d}x=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\arcsin\frac{x}{a}+C.$

68. $\int\sqrt{(a^2-x^2)^3}\,\mathrm{d}x=\frac{x}{8}\left(5a^2-2x^2\right)\sqrt{a^2-x^2}+\frac{3}{8}a^4\arcsin\frac{x}{a}+C.$

69. $\int x\sqrt{a^2-x^2}\,\mathrm{d}x=-\frac{1}{3}\sqrt{(a^2-x^2)^3}+C.$

70. $\int x^2\sqrt{a^2-x^2}\,\mathrm{d}x=\frac{x}{8}\left(2x^2-a^2\right)\sqrt{a^2-x^2}+\frac{a^4}{8}\arcsin\frac{x}{a}+C.$

71. $\int\frac{\sqrt{a^2-x^2}}{x}\,\mathrm{d}x=\sqrt{a^2-x^2}+a\ln\frac{a-\sqrt{a^2-x^2}}{|x|}+C.$

72. $\int\frac{\sqrt{a^2-x^2}}{x^2}\,\mathrm{d}x=-\frac{\sqrt{a^2-x^2}}{x}-\arcsin\frac{x}{a}+C.$

(九) 含有 $\sqrt{\pm ax^2+bx+c}(a>0)$ 的积分

73. $\int \frac{\mathrm{d}x}{\sqrt{ax^2+bx+c}} = \frac{1}{\sqrt{a}} \ln\left|2ax+b+2\sqrt{a}\sqrt{ax^2+bx+c}\right| + C.$

74. $\int \sqrt{ax^2+bx+c}\ \mathrm{d}x = \frac{2ax+b}{4a}\sqrt{ax^2+bx+c} +$

$$\frac{4ac-b^2}{8\sqrt{a^3}} \ln\left|2ax+b+2\sqrt{a}\sqrt{ax^2+bx+c}\right| + C.$$

75. $\int \frac{x}{\sqrt{ax^2+bx+c}}\ \mathrm{d}x = \frac{1}{a}\sqrt{ax^2+bx+c} -$

$$\frac{b}{2\sqrt{a^3}} \ln\left|2ax+b+2\sqrt{a}\sqrt{ax^2+bx+c}\right| + C.$$

76. $\int \frac{\mathrm{d}x}{\sqrt{c+bx-ax^2}} = \frac{1}{\sqrt{a}} \arcsin\frac{2ax-b}{\sqrt{b^2+4ac}} + C.$

77. $\int \sqrt{c+bx-ax^2}\ \mathrm{d}x = \frac{2ax-b}{4a}\sqrt{c+bx-ax^2} +$

$$\frac{b^2+4ac}{8\sqrt{a^3}} \arcsin\frac{2ax-b}{\sqrt{b^2+4ac}} + C.$$

78. $\int \frac{x}{\sqrt{c+bx-ax^2}}\ \mathrm{d}x = \frac{1}{a}\sqrt{c+bx-ax^2} + \frac{b}{2\sqrt{a^3}} \arcsin\frac{2ax-b}{\sqrt{b^2+4ac}} + C.$ (+) 含有 $\sqrt{\pm\frac{x-a}{x-b}}$ 或 $\sqrt{(x-a)(b-x)}$ 的积分

79. $\int \sqrt{\frac{x-a}{x-b}}\ \mathrm{d}x = (x-b)\sqrt{\frac{x-a}{x-b}} + (b-a)\ln(\sqrt{|x-a|} + \sqrt{|x-b|}) + C.$

80. $\int \sqrt{\frac{x-a}{b-x}}\ \mathrm{d}x = (x-b)\sqrt{\frac{x-a}{b-x}} + (b-a)\arcsin\sqrt{\frac{x-a}{b-a}} + C.$

81. $\int \frac{\mathrm{d}x}{\sqrt{(x-a)(b-x)}} = 2\arcsin\sqrt{\frac{x-a}{b-a}} + C(a<b).$

82. $\int \sqrt{(x-a)(b-x)}\mathrm{d}x = \frac{2x-a-b}{4}\sqrt{(x-a)(b-x)} +$

$$\frac{(b-a)^2}{4}\arcsin\sqrt{\frac{x-a}{b-a}} + C(a<b).$$

(十一) 含有三角函数的积分

83. $\int \sin x\ \mathrm{d}x = -\cos x + C.$

84. $\int \cos x\ \mathrm{d}x = \sin x + C.$

85. $\int \tan x\ \mathrm{d}x = -\ln|\cos x| + C.$

86. $\int \cot x\ \mathrm{d}x = \ln|\sin x| + C.$

87. $\int \sec x dx = \ln\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right| + C = \ln|\sec x + \tan x| + C.$

88. $\int \csc x \ \mathrm{d}x = \ln\left|\tan\frac{x}{2}\right| + C = \ln|\csc x - \cot x| + C.$

89. $\int \sec^2 x \ \mathrm{d}x = \tan x + C.$

90. $\int \csc^2 x \ \mathrm{d}x = -\cot x + C.$

91. $\int \sec x \tan x \ \mathrm{d}x = \sec x + C.$

92. $\int \csc x \cot x \ \mathrm{d}x = -\csc x + C.$

93. $\int \sin^2 x \ \mathrm{d}x = \frac{x}{2} - \frac{1}{4}\sin 2x + C.$

94. $\int \cos^2 x \ \mathrm{d}x = \frac{x}{2} + \frac{1}{4}\sin 2x + C.$

95. $\int \sin^n x \ \mathrm{d}x = -\frac{1}{n}\sin^{n-1} x \cos x + \frac{n-1}{n}\int \sin^{n-2} x \ \mathrm{d}x.$

96. $\int \cos^n x \ \mathrm{d}x = \frac{1}{n}\cos^{n-1} x \sin x + \frac{n-1}{n}\int \cos^{n-2} x \ \mathrm{d}x.$

97. $\int \frac{\mathrm{d}x}{\sin^n x} = -\frac{1}{n-1}\cdot\frac{\cos x}{\sin^{n-1} x} + \frac{n-2}{n-1}\int \frac{\mathrm{d}x}{\sin^{n-2} x}.$

98. $\int \frac{\mathrm{d}x}{\cos^n x} = \frac{1}{n-1}\cdot\frac{\sin x}{\cos^{n-1} x} + \frac{n-2}{n-1}\int \frac{\mathrm{d}x}{\cos^{n-2} x}.$

99. $\int \cos^m x \sin^* x \ \mathrm{d}x = \frac{1}{m+n}\cos^{m-1} x \sin^{n+1} x + \frac{m-1}{m+n}\int \cos^{m-2} x \sin^* x \ \mathrm{d}x =$
$\frac{1}{m+n}\cos^{m+1} x \sin^{n-1} x + \frac{n-1}{m+n}\int \cos^m x \sin^{n-2} x \ \mathrm{d}x.$

100. $\int \sin ax \cos bx \ \mathrm{d}x = -\frac{1}{2(a+b)}\cos(a+b)x - \frac{1}{2(a-b)}\cos(a-b)x + C.$

101. $\int \sin ax \sin bx \ \mathrm{d}x = \frac{1}{2(a+b)}\sin(a+b)x + \frac{1}{2(a-b)}\sin(a-b)x + C.$

102. $\int \cos ax \cos bx \ \mathrm{d}x = \frac{1}{2(a+b)}\sin(a+b)x + \frac{1}{2(a-b)}\sin(a-b)x + C.$

103. $\int \frac{\mathrm{d}x}{a+b\sin x} = \frac{2}{\sqrt{a^2-b^2}}\arctan\frac{\operatorname{atan}\frac{x}{2}+b}{\sqrt{a^2-b^2}} + C\left(a^2 > b^2\right).$

104. $\int \frac{\mathrm{d}x}{a+b\sin x} = \frac{1}{\sqrt{b^2-a^2}}\ln\left|\frac{a\tan\frac{x}{2}+b-\sqrt{b^2-a^2}}{a\tan\frac{x}{2}+b+\sqrt{b^2-a^2}}\right| + C\left(a^2 < b^2\right).$

105. $\int \frac{\mathrm{d}x}{a+b\cos x} = \frac{2}{a+b}\sqrt{\frac{a+b}{a-b}}\arctan\left(\sqrt{\frac{a-b}{a+b}}\tan\frac{x}{2}\right) + C\left(a^2 > b^2\right).$

106. $\int \frac{\mathrm{d}x}{a+b\cos x} = \frac{1}{a+b}\sqrt{\frac{a+b}{b-a}}\ln\left|\frac{\tan\frac{x}{2}+\sqrt{\frac{a+b}{b-a}}}{\tan\frac{x}{2}-\sqrt{\frac{a+b}{b-a}}}\right| + C\ (a^2 < b^2)$.

107. $\int \frac{\mathrm{d}x}{a^2\cos^2 x + b^2\sin^2 x} = \frac{1}{ab}\arctan\left(\frac{b}{a}\tan x\right) + C$.

108. $\int \frac{\mathrm{d}x}{a^2\cos^2 x - b^2\sin^2 x} = \frac{1}{2ab}\ln\left|\frac{b\tan x + a}{b\tan x - a}\right| + C$.

109. $\int x\sin ax\ \mathrm{d}x = \frac{1}{a^2}\sin ax - \frac{1}{a}x\cos ax + C$.

110. $\int x^2\sin ax\ \mathrm{d}x = -\frac{1}{a}x^2\cos ax + \frac{2}{a^2}x\sin ax + \frac{2}{a^3}\cos ax + C$.

111. $\int x\cos ax\ \mathrm{d}x = \frac{1}{a^2}\cos ax + \frac{1}{a}x\sin ax + C$.

112. $\int x^2\cos ax\ \mathrm{d}x = \frac{1}{a}x^2\sin ax + \frac{2}{a^2}x\cos ax - \frac{2}{a^3}\sin ax + C$.

(十二) 含有反三角函数的积分（其中 $a > 0$）

113. $\int \arcsin\frac{x}{a}\ \mathrm{d}x = x\arcsin\frac{x}{a} + \sqrt{a^2 - x^2} + C$.

114. $\int x\arcsin\frac{x}{a}\ \mathrm{d}x = \left(\frac{x^2}{2} - \frac{a^2}{4}\right)\arcsin\frac{x}{a} + \frac{x}{4}\sqrt{a^2 - x^2} + C$.

115. $\int x^2\arcsin\frac{x}{a}\ \mathrm{d}x = \frac{x^3}{3}\arcsin\frac{x}{a} + \frac{1}{9}\left(x^2 + 2a^2\right)\sqrt{a^2 - x^2} + C$.

116. $\int \arccos\frac{x}{a}\ \mathrm{d}x = x\arccos\frac{x}{a} - \sqrt{a^2 - x^2} + C$.

117. $\int x\arccos\frac{x}{a}\ \mathrm{d}x = \left(\frac{x^2}{2} - \frac{a^2}{4}\right)\arccos\frac{x}{a} - \frac{x}{4}\sqrt{a^2 - x^2} + C$.

118. $\int x^2\arccos\frac{x}{a}\ \mathrm{d}x = \frac{x^3}{3}\arccos\frac{x}{a} - \frac{1}{9}\left(x^2 + 2a^2\right)\sqrt{a^2 - x^2} + C$.

119. $\int \arctan\frac{x}{a}\ \mathrm{d}x = x\arctan\frac{x}{a} - \frac{a}{2}\ln\left(a^2 + x^2\right) + C$.

120. $\int x\arctan\frac{x}{a}\ \mathrm{d}x = \frac{1}{2}\left(a^2 + x^2\right)\arctan\frac{x}{a} - \frac{a}{2}x + C$.

121. $\int x^2\arctan\frac{x}{a}\ \mathrm{d}x = \frac{x^3}{3}\arctan\frac{x}{a} - \frac{a}{6}x^2 + \frac{a^3}{6}\ln\left(a^2 + x^2\right) + C$.

(十三) 含有指数函数的积分

122. $\int a^*\ \mathrm{d}x = \frac{1}{\ln a}a^* + C$.

123. $\int \mathrm{e}^{*x}\ \mathrm{d}x = \frac{1}{a}\mathrm{e}^{ax} + C$.

124. $\int x\mathrm{e}^{ax}\ \mathrm{d}x = \frac{1}{a^2}(ax - 1)\mathrm{e}^{ax} + C$.

125. $\int x^n \mathrm{e}^{ax}\ \mathrm{d}x = \frac{1}{a}x^n \mathrm{e}^{ax} - \frac{n}{a}\int x^{n-1}\mathrm{e}^{ax}\ \mathrm{d}x.$

126. $\int xa^*\ \mathrm{d}x = \frac{x}{\ln a}a^* - \frac{1}{(\ln a)^2}a^* + C.$

127. $\int x^n a^*\ \mathrm{d}x = \frac{1}{\ln a}x^n a^x - \frac{n}{\ln a}\int x^{n-1}a^2\ \mathrm{d}x.$

128. $\int \mathrm{e}^{ax}\sin bx\ \mathrm{d}x = \frac{1}{a^2+b^2}\mathrm{e}^{-a}(a\sin bx - b\cos bx) + C.$

129. $\int e^{ax}\cos bx\ \mathrm{d}x = \frac{1}{a^2+b^2}\mathrm{e}^{ax}(b\sin bx + a\cos bx) + C.$

130. $\int \mathrm{e}^{-x}\sin^* bx\ \mathrm{d}x = \frac{1}{a^2+b^2n^2}\mathrm{e}^{ax}\sin^{*-1} bx(a\sin bx - nb\cos bx) + \frac{n(n-1)b^2}{a^2+b^2n^2}\int \mathrm{e}^{ex}\sin^{n-2} bx\ \mathrm{d}x.$

131.
$$\int \mathrm{e}^{an}\cos^n bx\ \mathrm{d}x = \frac{1}{a^2+b^2n^2}\mathrm{e}^{2x}\cos^{-1} bx(a\cos bx + nb\sin bx) + \frac{n(n-1)b^2}{a^2+b^2n^2}\int \mathrm{e}^{-x}\cos^{-2-2} bx\ \mathrm{d}x$$

132. $\int \ln x\ \mathrm{d}x = x\ln x - x + C.$

133. $\int \frac{\mathrm{d}x}{x\ln x} = \ln|\ln x| + C.$

134. $\int x^*\ln x\ \mathrm{d}x = \frac{1}{n+1}x^{*+1}\left(\ln x - \frac{1}{n+1}\right) + C.$

135. $\int (\ln x)^n\ \mathrm{d}x = x(\ln x)^* - n\int (\ln x)^{n-1}\ \mathrm{d}x.$

136. $\int x^n(\ln x)^n\ \mathrm{d}x = \frac{1}{m+1}x^{m+1}(\ln x)^n - \frac{n}{m+1}\int x^m(\ln x)^{m-1}\ \mathrm{d}x.$

(十五) 含有双曲函数的积分

137. $\int \mathrm{sh}\, x\ \mathrm{d}x = \mathrm{ch}\, x + C.$

138. $\int \mathrm{ch}\, x\ \mathrm{d}x = \mathrm{sh}\, x + C.$

139. $\int \mathrm{th}\ x\ \mathrm{d}x = \ln \mathrm{ch}\, x + C.$

140. $\int \mathrm{sh}^2 x\ \mathrm{d}x = -\frac{x}{2} + \frac{1}{4}\mathrm{sh}\, 2x + C.$

141. $\int \mathrm{ch}^2 x\ \mathrm{d}x = \frac{x}{2} + \frac{1}{4}\mathrm{sh}\, 2x + C.$

(十六) 定积分

142. $\int_{-\infty}^{*} \cos nx\ \mathrm{d}x = \int_{-\infty}^{*} \sin nx\ \mathrm{d}x = 0.$

143. $\int_{-\infty}^{*} \cos mx \sin nx\ \mathrm{d}x = 0.$

144. $\int_{\cdots}^{*} \cos mx \cos nx \ \mathrm{d}x = \begin{cases} 0, m \neq n, \\ \pi, m = n. \end{cases}$

145. $\int_{\cdots}^{n} \sin mx \sin nx \ \mathrm{d}x = \begin{cases} 0, m \neq n. \\ \pi, m = n. \end{cases}$

146. $\int_0^{\pi} \sin mx \sin nx \ \mathrm{d}x = \int_0^{\pi} \cos mx \cos nx \ \mathrm{d}x = \begin{cases} 0, m \neq n, \\ \frac{\pi}{2}, m = n. \end{cases}$

147.

$$I_0 = \int_0^{\frac{\pi}{2}} \sin^x x \ \mathrm{d}x = \int_0^{\frac{\pi}{2}} \cos^* x \ \mathrm{d}x,$$

$$I_n = \frac{n-1}{n} I_{n-2}$$

$$= \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots\cdots \frac{4}{5} \cdot \frac{2}{3} (n \text{ 为大于1 的正奇数}), I_1 = 1, \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots\cdots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} (n \text{ 为正偶数}), I_0 = \frac{\pi}{2}. \end{cases}$$

I、ax+b

4、$\int \frac{x^2}{ax+b}\,dx = \int \frac{(ax+b)(\frac{x}{a}-\frac{b}{a^2})+\frac{b^2}{a^3}}{ax+b}\,dx = \int (\frac{x}{a}-\frac{b}{a^2})+\frac{b^2}{a^2}\frac{1}{ax+b}\,dx$

$= \frac{x^2}{2a} - \frac{b}{a^2}x + \frac{b^2}{a^3}\ln|ax+b| + C$

5. $\int \frac{1}{x(ax+b)}\,dx = \int \frac{1}{bx} - \frac{a}{b}\frac{1}{ax+b}\,dx = \frac{1}{b}\ln\left|\frac{x}{ax+b}\right| + C$

$\frac{A}{x}+\frac{B}{ax+b} = \frac{(aA+B)x+Ab}{x(ax+b)} \Rightarrow \begin{cases} Ab=1 \\ aA+B=0 \end{cases} \Rightarrow \begin{cases} A=\frac{1}{b} \\ B=-\frac{a}{b} \end{cases}$

6. $\int \frac{1}{x^2(ax+b)}\,dx = \int -\frac{a}{b^2}\frac{1}{x} + \frac{1}{bx^2} + \frac{a^2}{b^2}\frac{1}{ax+b}\,dx = -\frac{1}{bx} + \frac{a}{b^2}\ln\left|\frac{ax+b}{x}\right| + C$

$\frac{A}{x}+\frac{B}{x^2}+\frac{C}{ax+b} \Rightarrow \begin{cases} Aa+C=0 \\ Ab+Ba=0 \\ Bb=1 \end{cases} \Rightarrow \begin{cases} A=-\frac{a}{b^2} \\ B=\frac{1}{b} \\ C=\frac{a^2}{b^2} \end{cases}$

7. $\int \frac{x}{(ax+b)^2}\,dx = \int \frac{\frac{u-b}{a}}{u^2}\frac{du}{a} = \frac{1}{a^2}\int \frac{u-b}{u^2}\,du = \frac{1}{a^2}\ln|ax+b| + \frac{b}{a^2(ax+b)} + C$

令 $u=ax+b$, $x=\frac{u-b}{a}$

8. $\int \frac{x^2}{(ax+b)^2}\,dx = \int \frac{(\frac{u-b}{a})^2}{u^2}\frac{du}{a} = \frac{1}{a^3}\int \frac{u^2-2bu+b^2}{u^2}\,du$

$= \frac{1}{a^3}\left(ax+b-2b\ln|ax+b| - \frac{b^2}{ax+b}\right) + C$

9. $\int \frac{dx}{x(ax+b)^2} = \frac{1}{b^2}\int \frac{1}{x}dx - \frac{a}{b^2}\int \frac{1}{ax+b}\,dx - \frac{a}{b}\int \frac{1}{(ax+b)^2}\,dx$

$\frac{A}{x}+\frac{B}{ax+b}+\frac{C}{(ax+b)^2} \Rightarrow \begin{cases} Aa^2+Ba=0 \\ 2Aab+Bb+C=0 \\ Ab^2=1 \end{cases} \Rightarrow \begin{cases} A=\frac{1}{b^2} \\ B=-\frac{a}{b^2} \\ C=-\frac{a}{b} \end{cases}$

$= \frac{1}{b^2}\ln\left|\frac{x}{ax+b}\right| + \frac{1}{b(ax+b)} + C$

Ⅱ $\sqrt{ax+b}$

11. $\int x\sqrt{ax+b}\,dx$ 令 $u=ax+b$，$x=\frac{u-b}{a}$

$=\int \frac{u-b}{a}\sqrt{u}\,\frac{du}{a}=\frac{1}{a^2}\int u^{\frac{3}{2}}-bu^{\frac{1}{2}}\,du=\frac{2}{5a^2}u^{\frac{5}{2}}-\frac{2b}{3a^2}u^{\frac{3}{2}}+C$

$=\frac{2}{a^2}u^{\frac{3}{2}}\left(\frac{u}{5}-\frac{b}{3}\right)+C=\frac{2}{15a^2}(ax+b)^{\frac{3}{2}}(3ax-2b)$

12. $\int x^2\sqrt{ax+b}\,dx=\int\frac{(u-b)^2}{a^2}\sqrt{u}\,\frac{du}{a}=\frac{1}{a^3}\int u^{\frac{5}{2}}-2bu^{\frac{3}{2}}+b^2u^{\frac{1}{2}}\,du$

$=\frac{2}{7a^3}(ax+b)^{\frac{7}{2}}-\frac{4b}{5}(ax+b)^{\frac{5}{2}}+\frac{2b^2}{3}(ax+b)^{\frac{3}{2}}+C$

13. $\int\frac{x}{\sqrt{ax+b}}dx=\int\frac{\frac{u-b}{a}}{\sqrt{u}}\frac{du}{a}=\frac{1}{a^2}\int\sqrt{u}-\frac{b}{\sqrt{u}}du$

$=\frac{2}{3a^2}(ax+b)^{\frac{3}{2}}-\frac{2b}{a^2}\sqrt{ax+b}+C$

14. $\int\frac{x^2}{\sqrt{ax+b}}dx=\int\frac{(\frac{u-b}{a})^2}{\sqrt{u}}\frac{du}{a}=\frac{1}{a^3}\int u^{\frac{3}{2}}-2bu^{\frac{1}{2}}+\frac{b^2}{\sqrt{u}}du$

$=\frac{2}{5a^3}(ax+b)^{\frac{5}{2}}-\frac{4b}{3a^3}(ax+b)^{\frac{3}{2}}+\frac{2b^2}{a^3}(ax+b)^{\frac{1}{2}}+C$

15. $\int\frac{1}{x\sqrt{ax+b}}dx$ 令 $u=\sqrt{ax+b}$，$x=\frac{u^2-b}{a}$ $=\int\frac{a}{u(u^2-b)}\frac{2u}{a}du=2\int\frac{1}{u^2-b}du$

① 当 $b<0$ 时，$2\int\frac{1}{u^2+k}du=\frac{2}{\sqrt{k}}\arctan\left(\frac{u}{\sqrt{k}}\right)+C=\frac{2}{\sqrt{-b}}\arctan\left(\sqrt{\frac{ax+b}{-b}}\right)+C$

② 当 $b=0$ 时，$2\int\frac{1}{u^2}du=-\frac{2}{u}+C=-\frac{2}{\sqrt{ax}}+C$

③ 当 $b>0$ 时，$2\int\frac{1}{u^2-b}du=\frac{2}{\sqrt{b}}\int\frac{1}{u-\sqrt{b}}-\frac{1}{u+\sqrt{b}}du=\frac{1}{\sqrt{b}}\ln\left|\frac{\sqrt{ax+b}-\sqrt{b}}{\sqrt{ax+b}+\sqrt{b}}\right|+C$

16. $\int\frac{dx}{x^2\sqrt{ax+b}}=\int\frac{1}{(\frac{u^2-b}{a})^2\cdot u}\frac{2u}{a}du=2a\int\frac{du}{(u^2-b)^2}$

$=\frac{-u}{2a^2(u^2-b)^2}-\frac{1}{2a^2}\int\frac{du}{u^2-a^2}=-\frac{\sqrt{ax+b}}{bx}-\frac{a}{2b}\int\frac{dx}{x\sqrt{ax+b}}$

分部积分法

17. $\int \frac{\sqrt{ax+b}}{x}dx = \int \frac{au}{u^2-b}\frac{2u}{a}du = \int \frac{2u^2}{u^2-b}du = 2\sqrt{ax+b} + b\int \frac{1}{x\sqrt{ax+b}}dx$

18. $\int \frac{\sqrt{ax+b}}{x^2}dx = \int \sqrt{ax+b}\,d(-\frac{1}{x}) = -\frac{\sqrt{ax+b}}{x} + \frac{a}{2}\int \frac{dx}{x\sqrt{ax+b}}$

Ⅲ $x^2 \pm a^2$

20. $\int \frac{dx}{(x^2+a^2)^n}$

当 $n=1$ 时, $I_1 = \int \frac{dx}{x^2+a^2} = \frac{1}{a}\arctan(\frac{x}{a}) + C$

对 $I_{n-1} = \int \frac{dx}{(x^2+a^2)^{n-1}}$ 用分部积分. 设 $u = \frac{1}{(x^2+a^2)^{n-1}}$ $v = x$

则 $I_{n-1} = \frac{x}{(x^2+a^2)^{n-1}} - \int x\left(\frac{2(n-1)x}{(x^2+a^2)^n}\right)dx$

$= \frac{x}{(x^2+a^2)^{n-1}} + 2(n-1)\int \frac{x^2}{(x^2+a^2)^n}dx$ $= \int \frac{dx}{(x^2+a^2)^{n-1}} - a^2\int \frac{dx}{(x^2+a^2)^n}$

$= I_{n-1} - a^2 I_n$

$= \frac{1}{(x^2+a^2)^{n-1}} + 2(n-1)(I_{n-1} - a^2 I_n)$

$\Rightarrow I_n = \frac{1}{2(n-1)a^2}\left(\frac{x}{(x^2+a^2)^{n-1}} + (2n-3)I_{n-1}\right)$

IV ax^2+b

22. $$\int \frac{dx}{ax^2+b} = \begin{cases} \frac{1}{\sqrt{ab}} \arctan\sqrt{\frac{a}{b}}x + C & (b>0) \\ \frac{1}{2\sqrt{-ab}} \ln\left|\frac{\sqrt{a}x-\sqrt{-b}}{\sqrt{a}x+\sqrt{-b}}\right| + C & (b<0) \end{cases}$$

23. $$\int \frac{x}{ax^2+b}\,dx = \frac{1}{2a}\int \frac{1}{ax^2+b}\,dax^2 = \frac{1}{2a}\ln|ax^2+b| + C$$

25. $$\int \frac{dx}{x(ax^2+b)} = \int \frac{1}{bx} - \frac{a}{b}\frac{x}{ax^2+b}\,dx = \frac{1}{b}\ln|x| - \frac{1}{2b}\ln|ax^2+b| + C$$

$$\frac{A}{x} + \frac{Bx+C}{ax^2+b} \Rightarrow \begin{cases} Aa+B=0 \\ C=0 \\ Ab=1 \end{cases} \Rightarrow \begin{cases} A=\frac{1}{b} \\ B=-\frac{a}{b} \\ C=0 \end{cases}$$

27. $$\int \frac{dx}{x^3(ax^2+b)} = -\frac{1}{2}\int \frac{n\,du}{bu+a} = -\frac{1}{2bx^2} + \frac{a}{2b^2}\ln\left|\frac{b}{x^2}+a\right| + C$$

28. $$\int \frac{dx}{(ax^2+b)^n}$$

$$I_1 = \int \frac{dx}{ax^2+b} = \begin{cases} \frac{1}{\sqrt{ab}} \arctan\sqrt{\frac{a}{b}}x + C & (b>0) \\ \frac{1}{2\sqrt{-ab}} \ln\left|\frac{\sqrt{a}x-\sqrt{-b}}{\sqrt{a}x+\sqrt{-b}}\right| + C & (b<0) \end{cases}$$

$$I_{n-1} = \int \frac{dx}{(ax^2+b)^{n-1}} = \frac{x}{(ax^2+b)^{n-1}} + 2(n-1)\int \frac{ax^2}{(ax^2+b)^n}\,dx$$

$$= \frac{x}{(ax^2+b)^{n-1}} + 2(n-1)(I_{n-1} - bI_n)$$

$$\Rightarrow I_n = \frac{x}{2b(n-1)(ax^2+b)^{n-1}} + \frac{2n-3}{2b(n-1)} I_{n-1}$$

$n=2$ 时 $$I_2 = \frac{x}{2b(ax^2+b)} + \frac{1}{2b} I_1 = \frac{x}{2b(ax^2+b)} + \frac{1}{2b}\int \frac{dx}{ax^2+b}$$

V. ax^2+bx+c

29. $$\int \frac{dx}{ax^2+bx+c} = \int \frac{dx}{a\left(x+\frac{b}{2a}\right)^2 - \frac{b^2-4ac}{4a}}$$

$$= \begin{cases} \frac{2}{\sqrt{4ac-b^2}} \arctan \frac{2ax+b}{\sqrt{4ac-b^2}} + C & (b^2<4ac) \\ \frac{1}{\sqrt{b^2-4ac}} \ln \left| \frac{2ax+b-\sqrt{b^2-4ac}}{2ax+b+\sqrt{b^2-4ac}} \right| + C & (b^2>4ac) \end{cases}$$

30. $$\int \frac{x}{ax^2+bx+c}\,dx = \int \frac{\left(x+\frac{b}{2a}\right)dx}{a\left(x+\frac{b}{2a}\right)^2 - \frac{b^2-4ac}{4a}} - \frac{b}{2a}\int \frac{dx}{ax^2+bx+c}$$

$$= \frac{1}{2}\int \frac{d\left(x+\frac{b}{2a}\right)^2}{a\left(x+\frac{b}{2a}\right)^2 - \frac{b^2-4ac}{4a}} - \frac{b}{2a}\int \frac{dx}{ax^2+bx+c}$$

$$= \frac{1}{2a}\ln\left|ax^2+bx+c\right| - \frac{b}{2a}\int \frac{dx}{ax^2+bx+c} + C$$

VI. $\sqrt{x^2+a^2}$

1. 双曲变换：

$\sinh x = \frac{e^x - e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$　　$\cosh x = \frac{e^x + e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

$\sin x = \frac{e^{ix} - e^{-ix}}{2} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$　　$\cos x = \frac{e^{ix} + e^{-ix}}{2} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$

$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$　　$\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

$\tan x = \frac{e^{ix} - e^{-ix}}{e^{ix} + e^{-ix}}$　　$\coth x = \frac{e^{ix} + e^{-ix}}{e^{ix} - e^{-ix}}$

核心等式：

① $\cos^2 x + \sin^2 x = 1$　　$\cosh^2 x - \sinh^2 x = 1$

② $\sin 2x = 2\sin x \cos x$　　$\sinh 2x = 2\sinh x \cosh x$

$\cos 2x = \cos^2 x - \sin^2 x$　　$\cosh 2x = \cosh^2 x + \sinh^2 x$

③ $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$　　$\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$

$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$　　$\cos(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$

④ $\cos^2 x = \frac{1 + \cos 2x}{2}$　　$\cosh^2 x = \frac{\cosh 2x + 1}{2}$

$\sin^2 x = \frac{1 - \cos 2x}{2}$　　$\sinh^2 x = \frac{\cosh 2x - 1}{2}$

一般情况下，当出现 $\sqrt{a^2 - x^2}$ 时，使用三角代换。

当出现 $\sqrt{x^2 + a^2}$ 时，使用双曲代换

当出现 $(x^2 + a^2)$ 时，使用换元 / 分部 / 递推

导数和积分：

$\sinh' x = \cosh x$	$\sin' x = \cos x$
$\cosh' x = \sinh x$	$\cos' x = \sin x$
$\tanh' x = \operatorname{sech}^2 x = 1 - \tanh^2 x$	$\tan' x = \sec^2 x = 1 + \tan^2 x$
$\coth' x = -\operatorname{csch}^2 x = 1 - \coth^2 x$	$\cot' x = -\csc^2 x = -(1 + \cot^2 x)$
$\operatorname{sech}' x = -\operatorname{sech} x \tanh x$	$\sec' x = \sec x \tan x$
$\operatorname{csch}' x = -\operatorname{csch} x \coth x$	$\csc' x = -\csc x \cot x$

$\int \sinh x\,dx = \cosh x + C$	$\int \sin x\,dx = -\cos x + C$
$\int \cosh x\,dx = \sinh x + C$	$\int \cos x\,dx = \sin x + C$
$\int \tanh x\,dx = \ln(\cosh x) + C$	$\int \tan x\,dx = -\ln\lvert\cos x\rvert + C$
$\int \coth x\,dx = \ln\lvert\sinh x\rvert + C$	$\int \cot x\,dx = \ln\lvert\sin x\rvert + C$
$\int \operatorname{sech} x\,dx = 2\arctan e^x + C$ $= \arctan(\sinh x) + C$	$\int \sec x\,dx = \ln\lvert\sec x + \tan x\rvert + C$
$\int \operatorname{csch} x\,dx = \ln\lvert\tanh\frac{x}{2}\rvert + C$ $= -\ln\lvert\operatorname{csch} x + \coth x\rvert + C$	$\int \csc x\,dx = -\ln\lvert\csc x + \cot x\rvert + C$ $= \ln\lvert\tan\frac{x}{2}\rvert + C$
$\int \operatorname{sech}^2 x\,dx = \tanh x + C$	$\int \sec^2 x\,dx = \tan x + C$
$\int \operatorname{csch}^2 x\,dx = -\coth x + C$	$\int \csc^2 x\,dx = -\cot x + C$
$\int \operatorname{sech} x \tanh x\,dx = -\operatorname{sech} x + C$	$\int \sec x \tan x\,dx = \sec x + C$
$\int \operatorname{csch} x \coth x\,dx = -\operatorname{csch} x + C$	$\int \csc x \cot x\,dx = -\csc x + C$

反函数=

$\operatorname{arcsinh}' x = \frac{1}{\sqrt{1+x^2}}$　　$\arcsin' x = \frac{1}{\sqrt{1+x^2}}$

$\operatorname{arccosh}' x = \frac{1}{\sqrt{x^2-1}}$　　$\arccos' x = \frac{1}{\sqrt{x^2-1}}$

$\operatorname{arctanh}' x = \frac{1}{1-x^2}$ $|x|<1$　　$\arctan' x = \frac{1}{1-x^2}$

$\operatorname{arccoth}' x = \frac{1}{1-x^2}$ $|x|>1$　　$\operatorname{arccot}' x = \frac{1}{1-x^2}$

$\operatorname{arcsech}' x = \frac{1}{x\sqrt{1-x^2}}$　　$\operatorname{arcsec}' x = -\frac{1}{x\sqrt{1-x^2}}$

$\operatorname{arccsch}' x = \frac{1}{|x|\sqrt{1+x^2}}$　　$\operatorname{arccsc}' x = \frac{1}{|x|\sqrt{1+x^2}}$

反双曲函数的对数形式=

1. $\operatorname{arcsinh} x$

$y = \operatorname{arcsinh} x$，$x = \sinh y = \frac{e^y - e^{-y}}{2}$，令 $u = e^y > 0$，则

$x = \frac{u - \frac{1}{u}}{2} \Rightarrow u^2 - 2xu - 1 = 0 \Rightarrow u = x + \sqrt{x^2+1}$（舍去负值）

$\therefore e^y = x + \sqrt{x^2+1}$　　$y = \ln(x^2 + \sqrt{x^2+1})$

同理可得 $\operatorname{arccosh} x = \ln(x^2 + \sqrt{x^2-1})$

2. $\operatorname{arctanh} x$

$y = \operatorname{arctanh} x$，$x = \tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}}$，令 $u = e^y > 0$，则

$x = \frac{u - \frac{1}{u}}{u + \frac{1}{u}} = \frac{u^2-1}{u^2+1} \Rightarrow u^2 = \frac{1+x}{1-x} \Rightarrow u = \sqrt{\frac{1+x}{1-x}} = e^y$

$\therefore y = \ln\left(\sqrt{\frac{1+x}{1-x}}\right) = \frac{1}{2}(\ln(1+x) - \ln(1-x))$

分部+递推

$$\int \sinh^n x\,dx = \frac{1}{n}\sinh^{n-1}x\cosh x - \frac{n-1}{n}\int \sinh^{n-2}x\,dx,\quad n\geq 2$$

$$\int \sinh^n x\,dx = -\frac{1}{n}\sin^{n-1}x\cos x + \frac{n-1}{n}\int \sin^{n-2}x\,dx \quad n\geq 2$$

$$\int \cosh^n x\,dx = \frac{1}{n}\cosh^{n-1}x\sinh x + \frac{n-1}{n}\int \cosh^{n-2}x\,dx$$

$$\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1}x\sin x + \frac{n-1}{n}\int \cos^{n-2}x\,dx$$

31. $\int \frac{dx}{\sqrt{x^2+a^2}}$

$x = a\sinh t$, $dx = a\cosh t\, dt$

$$= \int \frac{a\cosh t\, dt}{\sqrt{a^2\sinh^2 t + a^2}} = t + C = \operatorname{arcsinh}\left(\frac{x}{a}\right) + C$$

$$= \ln\left(\frac{x}{a} + \sqrt{\frac{x^2}{a^2}+1}\right) + C = \ln\left(\frac{x+\sqrt{x^2+a^2}}{a}\right) + C$$

$x = a\tan\theta$, $dx = a\sec^2\theta\, d\theta$

$$= \int \frac{a}{\cos^2\theta}\, d\theta \Big/ \sqrt{a^2\tan^2\theta + a^2} = \int \frac{d\theta}{\cos\theta}$$

$$= \ln|\sec\theta + \tan\theta| + C = \ln\left(\frac{x+\sqrt{x^2+a^2}}{a}\right) + C$$

32. $\int \frac{dx}{\sqrt{(x^2+a^2)^3}}$

$x = a\sinh t$, $dx = a\cosh t$

$$= \int \frac{dt}{a^2\cosh^2 t} = \frac{1}{a^2}\tanh t + C$$

$$\tanh t = \frac{\sinh t}{\cosh t} = \frac{\frac{x}{a}}{\sqrt{x^2+a^2}/a} = \frac{x}{\sqrt{x^2+a^2}}$$

$x = a\tan\theta$, $dx = a\sec^2\theta\, d\theta$

$$= \frac{1}{a^2}\int \cos\theta\, d\theta = \frac{1}{a^2}\sin\theta + C$$

$$\tan\theta = \frac{x}{a} \Rightarrow \sin\theta = \frac{x}{\sqrt{x^2+a^2}}$$

25. $\int \frac{x^2}{\sqrt{x^2+a^2}}dx =$

令 $x = a\sinh t$

$dx = a\cosh t\,dt$

$$= \int \frac{a^2\sinh^2 t\cosh t}{a\cosh t}dt = a^2\int \sinh^2 t\,dt$$

$$= a^2\int \frac{\cosh 2t - 1}{2}dt = \frac{a^2}{2}\left(\frac{\sinh 2t}{2} - t\right) + C$$

$$= \frac{x}{2}\sqrt{x^2+a^2} - \frac{a^2}{2}\ln\left(x+\sqrt{x^2+a^2}\right) + C'$$

分部积分法

$$= \int x\frac{x}{\sqrt{x^2+a^2}}dx = \int x\,d\sqrt{x^2+a^2}$$

$$= x\sqrt{x^2+a^2} - \int \sqrt{x^2+a^2}\,dx$$

令 $x = a\sinh t$

$dx = a\cosh t\,dt$

$$\int \sqrt{x^2+a^2}\,dx = a^2\int \cosh^2 t\,dt = \frac{a^2}{2}\int 1+\cosh 2t\,dt$$

$$= \frac{a^2}{2}t + \frac{a^2}{4}\sinh 2t + C = \frac{a^2}{2}t + \frac{a^2}{2}\sinh t\cosh t + C$$

$$= \frac{a^2}{2}\ln\left(x+\sqrt{x^2+a^2}\right) + \frac{x}{2}\sqrt{x^2+a^2} + C$$

代入得原积分 $= \frac{x}{2}\sqrt{x^2+a^2} - \frac{a^2}{2}\ln\left(x+\sqrt{x^2+a^2}\right) + C$

VII. $\sqrt{x^2-a^2}$

43. $\int \sqrt{x^2-a^2}\,dx$

令 $x=a\cosh t$, $dx=a\sinh t\,dt$

$$= \int a^2\sinh^2 t\,dt = a^2\int \frac{\cosh 2t-1}{2}\,dt = \frac{a^2}{2}\left(\frac{\sinh 2t}{2}-t\right)+C$$

$$= \frac{a^2}{2}(\sinh t\cosh t - t)+C$$

$$= \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left(x+\sqrt{x^2-a^2}\right)+C'$$

45. $\int \frac{1}{\sqrt{x^2-a^2}}\,dx$

令 $x=a\cosh t$, $dx=a\sinh t\,dt$

$$= \int 1\,dt = t+C = \operatorname{arccosh}\frac{x}{a}+C$$

49. $\int \frac{x^2}{\sqrt{x^2-a^2}}\,dx = \int \frac{a^2\cosh^2 t\sinh t}{a\sinh t}\,dt = a^2\int \cosh^2 t = a^2\int \frac{\cosh 2t+1}{2}\,dt$

$$= \frac{a^2}{2}t + \frac{a^2}{4}\sinh 2t + C = \frac{a^2}{2}t + \frac{a^2}{2}\sinh t\cosh t + C$$

$t=\operatorname{arccosh}\frac{x}{a} = \ln\left|x+\sqrt{x^2-a^2}\right|$ 代入得

原积分 $= \frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right| + \frac{x}{2}\sqrt{x^2-a^2}+C$

57. $\int \frac{\sqrt{x^2-a^2}}{x}\,dx = \int \frac{a^2\sinh^2 t}{a\cosh t}\,dt = a\int \cosh t - \operatorname{sech} t\,dt$

$$= a\sinh t - a\arctan(\sinh t)+C$$

$= \arctan\left(\frac{\sqrt{x^2-a^2}}{a}\right)$ 设 $\tan\theta = \frac{\sqrt{x^2-a^2}}{a}$

则 $\cos\theta = \frac{a}{x}$, $\arctan\left(\frac{\sqrt{x^2-a^2}}{a}\right) = \arccos\left(\frac{a}{x}\right)$

$$= \sqrt{x^2-a^2} - a\arccos\left(\frac{a}{x}\right)+C$$

VIII. $\sqrt{a^2-x^2}$

59. $\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a} + C$

60. $\int \frac{dx}{\sqrt{(a^2-x^2)^3}} = \int \frac{a\cos t\,dt}{a^3\cos^3 t} = \frac{1}{a^2}\int \frac{dt}{\cos^2 t} = \frac{1}{a^2}\tan t + C$

$= \frac{1}{a^2}\frac{x}{\sqrt{a^2-x^2}} + C$

$\therefore x = a\sin t$

$dx = a\cos t\,dt$

63. $\int \frac{x^2}{\sqrt{a^2-x^2}}dx = \int a^2\sin^2 t\,dt = a^2\int \frac{1-\cos 2t}{2}dt$

$= \frac{a^2 t}{2} - \frac{a^2}{2}\sin t\cos t + C$

$= \frac{a^2}{2}\arcsin\frac{x}{a} - \frac{1}{2}x\sqrt{a^2-x^2} + C$

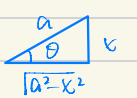
$\therefore x = a\sin t$

$dx = a\cos t\,dt$

64. $\int \frac{x^2}{\sqrt{(a^2-x^2)^3}}dx = \int \frac{a^2\sin^2 t\cos t}{a^3\cos^3 t}dt = \int \frac{\sin^2 t}{\cos^2 t}dt = \int \frac{1}{\cos^2 t} - 1\,dt$

$= \tan t - t + C = \frac{x}{\sqrt{a^2-x^2}} - \arcsin\frac{x}{a} + C$

$\tan\left(\arcsin\frac{x}{a}\right) = \frac{x}{\sqrt{a^2-x^2}}$



IX $\sqrt{ax^2+bx+C} = a(x+\frac{b}{2a})^2 - \frac{b^2-4ac}{4a}$

73 $\int \frac{dx}{\sqrt{ax^2+bx+C}} = \int \frac{dx}{\sqrt{a(x+\frac{b}{2a})^2 - \frac{b^2-4ac}{4a}}}$ 令 $u = x+\frac{b}{2a}$, $du = dx$ $= \int \frac{du}{\sqrt{au^2 - \frac{b^2-4ac}{4a}}}$

① 当 $a>0$ 时: $= \frac{1}{\sqrt{a}} \int \frac{du}{\sqrt{u^2 - \frac{\Delta}{4a^2}}}$

① 1. 当 $\Delta<0$ 时: $= \frac{1}{\sqrt{a}} \int \frac{du}{\sqrt{u^2+(-\frac{\sqrt{\Delta}}{2a})^2}} = \frac{1}{\sqrt{a}} \ln\left(x+\frac{b}{2a} + \frac{1}{\sqrt{a}}\sqrt{ax^2+bx+C}\right)+C$

2. 当 $\Delta>0$ 时: $= \frac{1}{\sqrt{a}} \int \frac{du}{\sqrt{u^2-(\frac{\sqrt{\Delta}}{2a})^2}} = \frac{1}{\sqrt{a}} \ln\left|x+\frac{b}{2a} + \frac{1}{\sqrt{a}}\sqrt{ax^2+bx+C}\right|+C$

② 当 $a<0$ 时:

则必有 $\Delta>0$ 且 x 介于两根之间, 使 $ax^2+bx+C>0$

$ax^2+bx+C = |a|\left(\frac{c}{|a|} + \frac{b}{|a|}x - x^2\right) = |a|\left(\frac{\Delta}{4a^2} - (x-\frac{b}{2|a|})^2\right)$

$= \frac{1}{\sqrt{|a|}} \int \frac{dx}{\sqrt{\frac{\Delta}{4a^2} - (x-\frac{b}{2|a|})^2}} = \frac{1}{\sqrt{|a|}} \int \frac{du}{\sqrt{\frac{\Delta}{4a^2}-u^2}} = \frac{1}{\sqrt{|a|}} \arcsin\left(\frac{2|a|u}{\sqrt{\Delta}}\right)+C$

$= \frac{1}{\sqrt{-a}} \arcsin\left(\frac{-2ax-b}{\sqrt{\Delta}}\right)+C$

74. $\int \sqrt{ax^2+bx+c}\,dx = \int \sqrt{a(x+\frac{b}{2a})^2 - \frac{b^2-4ac}{4a}}\,dx = \sqrt{a}\int \sqrt{u^2 - \frac{\Delta}{4a^2}}\,du$

① 当 $a>0$ 时: $= \frac{1}{2}(x+\frac{b}{2a})\sqrt{ax^2+bx+c} - \frac{\Delta\sqrt{a}}{8a^2}\ln(x+\frac{b}{2a}+\frac{1}{\sqrt{a}}\sqrt{ax^2+bx+c}) + C$

② 当 $a<0$ 时: $= \sqrt{a}\int \sqrt{\frac{\Delta}{4a^2} - (x+\frac{b}{2a})^2}\,dx$

$= \frac{1}{2}(x+\frac{b}{2a})\sqrt{ax^2+bx+c} + \frac{\Delta\sqrt{-a}}{8a^2}\arcsin\left(\frac{2ax+b}{\sqrt{b^2-4ac}}\right) + C$

75. $\int \frac{x}{\sqrt{ax^2+bx+c}}\,dx$

$\frac{d}{dx}\sqrt{ax^2+bx+c} = \frac{2ax+b}{2\sqrt{ax^2+bx+c}}$ 将 x 写成 $2ax+b$ 的线性组合 $x = \frac{1}{2a}(2ax+b) - \frac{b}{2a}$

$I = \frac{1}{2a}\int \frac{2ax+b}{\sqrt{ax^2+bx+c}}\,dx - \frac{b}{2a}\int \frac{1}{\sqrt{ax^2+bx+c}}\,dx$

$= \frac{1}{a}\sqrt{ax^2+bx+c} - \frac{b}{2a}\int \frac{1}{\sqrt{ax^2+bx+c}}\,dx + C$

① 当 $a>0$ 时: $= \frac{1}{a}\sqrt{ax^2+bx+c} - \frac{b}{2\sqrt{a^3}}\ln\left(x+\frac{b}{2a}+\frac{1}{\sqrt{a}}\sqrt{ax^2+bx+c}\right) + C$

② 当 $a<0$ 时: $= \frac{1}{a}\sqrt{ax^2+bx+c} - \frac{b}{2a\sqrt{-a}}\arcsin\left(\frac{-2ax-b}{\sqrt{\Delta}}\right) + C$

X $\sqrt{\frac{x-a}{x-b}}$ 或 $\sqrt{(x-a)(x-b)}$

79. $\int \sqrt{\frac{x-a}{x-b}}\,dx$ 令 $t=\sqrt{\frac{x-a}{x-b}}$, $x=\frac{bt^2-a}{t^2-1}$ $= \int t\,\frac{2t(a-b)}{(t^2-1)^2}\,dt$

$= 2(a-b)\int \frac{t^2}{(t^2-1)^2}\,dt = \frac{a-b}{2}\int \frac{1}{(t-1)^2}+\frac{1}{(t+1)^2}+\frac{1}{t-1}-\frac{1}{t-1}\,dt$

$= \frac{a-b}{2}\ln\left|\frac{t-1}{t+1}\right| - \frac{t(a-b)}{(t^2-1)}+C$

$= \frac{a-b}{2}\ln\left|\frac{\sqrt{x-a}-\sqrt{x-b}}{\sqrt{x-a}+\sqrt{x-b}}\right| + \sqrt{(x-a)(x-b)}+C$

$=(a-b)\left(\ln\left|\sqrt{x-a}-\sqrt{x-b}\right| - \frac{1}{2}\ln|b-a|\right)$

$= (a-b)\ln\left|\sqrt{x-a}-\sqrt{x-b}\right| + \sqrt{(x-a)(x-b)}+C$

or $= (b-a)\ln\left|\sqrt{x-a}+\sqrt{x-b}\right| + \sqrt{(x-a)(x-b)}+C$

80. $\int\sqrt{\frac{x-a}{b-x}}\,dx$ 令 $t=\sqrt{\frac{x-a}{b-x}}$, $x=\frac{bt^2+a}{1+t^2}$ $= \int \frac{2t^2(b-a)}{(1+t^2)^2}\,dt$

$= 2(b-a)\int \frac{t^2}{(1+t^2)^2} = 2(b-a)\left(\int\frac{dt}{1+t^2} - \int\frac{dt}{(1+t^2)^2}\right)$

$= 2(b-a)\left(\frac{1}{2}\arctan t - \frac{t}{2(1+t^2)}+C\right) = (b-a)\arctan t - \frac{(b-a)t}{1+t^2}+C$

$= (b-a)\arctan\sqrt{\frac{x-a}{b-x}} - \sqrt{(x-a)(b-x)}+C$

81. $\int \sqrt{(x-a)(x-b)}\,dx = \int \sqrt{x^2-(a+b)x+ab}\,dx = \int \sqrt{(x-\frac{a+b}{2})^2-(\frac{a-b}{2})^2}\,dx$

令 $u = x-\frac{a+b}{2}$

$k = |\frac{a-b}{2}|$

則 $(x-a)(x-b) = u^2-k^2$

$= \int \sqrt{u^2-k^2}\,du = \frac{u}{2}\sqrt{u^2-k^2} - \frac{k^2}{2}\ln\left|u+\sqrt{u^2-k^2}\right| + C$

$= \frac{2x-a-b}{4}\sqrt{(x-a)(x-b)} - \frac{(a-b)^2}{8}\ln\left|x-\frac{a+b}{2}+\sqrt{(x-a)(x-b)}\right| + C$

82. $\int \sqrt{(x-a)(b-x)}\,dx = \int \sqrt{k^2-u^2}\,du = \frac{u}{2}\sqrt{k^2-u^2} + \frac{k^2}{2}\arcsin\frac{u}{k} + C$

$= \frac{1}{2}(x-\frac{a+b}{2})\sqrt{(x-a)(b-x)} + \frac{(a-b)^2}{8}\arcsin\sqrt{\frac{x-a}{b-a}} + C$

83. $\int \frac{dx}{\sqrt{(x-a)(x-b)}} = \int \frac{dx}{\sqrt{u^2-k^2}} = \ln\left|u+\sqrt{u^2-k^2}\right| + C$

$= \ln\left|x-\frac{a+b}{2}+\sqrt{(x-a)(x-b)}\right| + C$

84. $\int \frac{dx}{\sqrt{(x-a)(b-x)}} = \int \frac{dx}{\sqrt{k^2-u^2}} = \arcsin\frac{u}{k} + C$

$= \arcsin\left(\frac{2x-a-b}{|a-b|}\right) + C$

XI 三角函数

95、$\int \sin^n dx = \int \sin^{n-1}x\, d(-\cos x)$

$= -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x\cos^2 x\, dx$ （$\cos^2 x = 1-\sin^2 x$）

$= -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x\, dx - (n-1)\int \sin^n dx$

$\Rightarrow n\int \sin^n dx = (n-1)\int \sin^{n-2}x\, dx - \sin^{n-1}x\cos x$

$$\int \sin^n dx = \frac{n-1}{n}\int \sin^{n-2}x\, dx - \frac{1}{n}\sin^{n-1}x\cos x$$

初始条件：$I_0 = \int dx = x + C$　　$I_1 = \int \sin x\, dx = -\cos x + C$

96、$\int \cos^n dx = \int \cos^{n-1}x\, d\sin x = \cos^{n-1}x\sin x + (n-1)\int \cos^{n-2}x\sin^2 x\, dx$

$= \cos^{n-1}x\sin x + (n-1)\int \cos^{n-2}x\, dx - (n-1)\int \cos^n dx$

$\Rightarrow n\int \cos^n dx = \cos^{n-1}x\sin x + (n-1)\int \cos^{n-2}x\, dx$

$$\int \cos^n dx = \frac{1}{n}\cos^{n-1}x\sin x + \frac{n-1}{n}\int \cos^{n-2}x\, dx$$

初始条件：$I_0 = \int dx = x + C$　　$I_1 = \int \cos x\, dx = \sin x + C$

97. $\int \frac{dx}{\sin^n x} = \int \csc^n dx = \int \csc^{n-2} x \, d(-\cot x)$

$= -\csc^n x \cot x - \int (n-2)\csc^{n-2} x \cot^2 x \, dx$ ($\csc^2 - 1$)

$\Rightarrow (n-1)\int \csc^n x \, dx = -\csc^{n-2} x \cot x + (n-2)\int \csc^{n-2} x$

$\int \csc^n x \, dx = -\frac{1}{n-1}\csc^{n-2} x \cot x + \frac{n-2}{n-2}\int \csc^{n-2} x$

初始条件 $= J_1 = \int \csc x \, dx = \ln|\csc x - \cot x| + C$

$J_2 = \int \csc^2 x \, dx = -\cot x + C$

98. $\int \frac{dx}{\cos^n x} = \int \sec^n x \, dx = \int \sec^{n-2} x \, d\tan x$

$= \sec^{n-2} x \tan x - (n-2)\int \sec^{n-2} x \tan^2 x \, dx$

$\Rightarrow (n-1)\int \sec^n x \, dx = \sec^{n-2} x \tan x - (n-2)\int \sec^{n-2} x \, dx$

$\int \sec^n x \, dx = \frac{1}{n-1}\sec^{n-2} x \tan x - \frac{n-2}{n-1}\int \sec^{n-2} x \, dx$

初始条件 $= S_1 = \int \sec x \, dx = \ln|\sec x + \tan x| + C$

$S_2 = \int \sec^2 x \, dx = \tan x + C$

103. $\int \frac{dx}{a+b\sin x}$ 令 $t=\tan\frac{x}{2}$, $\sin x=\frac{2t}{1+t^2}$, $dx=\frac{2}{1+t^2}dt$ $= \int \frac{1}{a+\frac{2bt}{1+t^2}} \frac{2}{1+t^2}dt = \int \frac{2}{at^2+2bt+a}dt$

代入得 $= \frac{2}{\sqrt{a^2-b^2}} \arctan \frac{a\tan\frac{x}{2}+b}{\sqrt{a^2+b^2}} + C \quad (a^2>b^2)$

$= \frac{1}{\sqrt{b^2-a^2}} \ln \left| \frac{a\tan\frac{x}{2}+b-\sqrt{b^2-a^2}}{a\tan\frac{x}{2}+b+\sqrt{b^2-a^2}} \right| + C \quad (a^2<b^2)$

105. $\int \frac{dx}{a+b\cos x}$ 令 $t=\tan\frac{x}{2}$, $\cos x=\frac{1-t^2}{1+t^2}$, $dx=\frac{2}{1+t^2}dt$ $= \int \frac{2}{(a-b)t^2+(a+b)}dt$

代入得 $= \frac{2}{\sqrt{a^2-b^2}} \arctan\left(\sqrt{\frac{a-b}{a+b}}\tan\frac{x}{2}\right) + C \quad (a^2>b^2)$

$= \frac{1}{\sqrt{b^2-a^2}} \ln \left| \frac{\tan\frac{x}{2}+\sqrt{\frac{a+b}{b-a}}}{\tan\frac{x}{2}-\sqrt{\frac{a+b}{b-a}}} \right| + C \quad (a^2<b^2)$

107. $$\int \frac{dx}{a^2\cos^2 x + b^2\sin^2 x} = \int \frac{\sec^2 x\,dx}{a^2+b^2\tan^2 x} = \int \frac{dt}{a^2+b^2t^2}$$
$$= \frac{1}{ab}\arctan\left(\frac{b}{a}t\right)+C = \frac{1}{ab}\arctan\left(\frac{b}{a}\tan x\right)+C$$

108. $$\int \frac{dx}{a^2\cos^2 x - b^2\sin^2 x} = \int \frac{\sec^2 x\,dx}{a^2 - b^2\tan^2 x} = \int \frac{dt}{a^2 - b^2 t}$$
$$= \frac{1}{2ab}\ln\left|\frac{bt+a}{bt-a}\right|+C = \frac{1}{2ab}\ln\left|\frac{b\tan x - a}{b\tan x + a}\right|+C$$

109. $$\int x\sin x\,dx = \int x\,d(-\cos x) = -x\cos x + \sin x + C$$
$$\int x\cos x\,dx = \int x\,d\sin x = x\sin x - \cos x + C$$

110. $$\int x^n \sin x\,dx = -x^n\cos x + n\int x^{n-1}\cos x\,dx$$
$$\int x^n \cos x\,dx = x^n\sin x - n\int x^{n-1}\sin x\,dx$$

設
$$\begin{cases} S_n = -x^n\cos x + nC_{n-1} = -x^n\cos x + nx^{n-1}\sin x - n(n-1)S_{n-2} \\ C_n = x^n\sin x - nS_{n-1} = x^n\sin x + nx^{n-1}\cos x - n(n-1)C_{n-2} \end{cases}$$

$S_0 = \int \sin x\,dx = -\cos x + C$ $\quad S_1 = -x\cos x + \sin x + C$

$C_0 = \int \cos x\,dx = \sin x + C$ $\quad C_1 = x\sin x + \cos x + C$

XII